

Adnan Sami Pavel

AI/ML Engineer (Signal Processing, Generative AI & Agentic AI)

+880 1884917523 adnansamipavel@gmail.com [linkedin.com](https://www.linkedin.com) github.com

Education

Green University of Bangladesh

Expected July 2025

Bachelor of Science in Computer Science and Engineering (CGPA: 3.14 / 4.00)

Dhaka, Bangladesh

- **Relevant Coursework:** Artificial Intelligence (Python), Machine Learning (Python), Data mining (Python), Data Structures and Algorithms (C++ & Java), OOP (Java), Database System (MySQL).

Research Projects

- **NeurIPS 2025 Challenge: Time-Series Denoising with Deep Learning** | *Pytorch*

- -

- Actively competing in a premier scientific challenge hosted in collaboration with NeurIPS, University College London (UCL), and the European Space Agency (ESA).
- Developing and training 1D CNN and Transformer models in PyTorch to remove cosmic ray noise from real-world planetary signal data.

- **Research & Publication: Structured Prompting and Multi-Agent Reasoning for Open-Source Financial Sentiment Analysis** | *python*

- -

- Co-authored a research paper proposing a novel framework for analyzing financial sentiment from statements using multi-agent architecture and open-source LLM models.
- The paper was successfully submitted to the IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS 2025) on July 31, 2025.
- My contributions included designing the core agentic logic, conducting experiments, and writing the Abstract and Methodology.

- **Independent Research: Foundational Language Model with State-Space Architecture** | *Pytorch*

-

- Conducting independent research into building a foundational Large Language Model from scratch in PyTorch, focusing on efficient, non-Transformer architectures like Mamba/S4.
- This ongoing project involves a deep dive into cutting-edge deep learning theory, model architecture design, and high-performance computing practices.

Technical Skills

AI/ML: PyTorch, Pandas, NumPy, Scikit-learn, Hugging Face, Matplotlib

Languages: Python, C++, Java, SQL, JavaScript

Core Concepts: Deep Learning, Large Language Models (LLMs), Agentic AI, Signal Processing, Transformers, CNNs, State-Space Models (SSM), Natural Language Processing (NLP), REST APIs

Frameworks & Tools: Git, GitHub, Docker, FastAPI, Flask, React.js, Node.js

Professional Experience

Intradoc International

November 1, 2024 – March 10, 2025

Front-End Developer & UI Designer

Dhaka, Bangladesh

- Collaborated in an agile team with developers, designers, and project managers to deliver projects on schedule.
- Used Git and GitHub for version control in a professional, collaborative environment.

QA Harbor Limited

March 20, 2025 – May 10, 2025

Software Quality Assurance Intern

Dhaka, Bangladesh

- Gained hands-on experience with the full Software Development Lifecycle (SDLC), including Agile models.
- Performed manual and automated testing using Playwright.
- Conducted API testing with Postman and applied both black-box and white-box testing techniques.